

HAOYU WANG

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🎓 EDUCATION

University of Texas at Austin, Austin, TX, USA

Jun. 2024–Present

Ph.D. in Electrical and Computer Engineering

- Graduate Research Assistant at *Power Electronics and Magnetics Group*
- Teaching Assistant at *ECE462 Power Electronics Lab*

Columbia University, New York, NY, USA

Aug. 2023–May 2024

Ph.D. in Electrical Engineering

- Teaching Assistant at *ELEN E4361 Power Electronics*

Tsinghua University, Beijing, China

Aug. 2020–Jun. 2023

M.S. in Electrical Engineering

- Thesis: *Efficiency Optimization and Decoupling Control for High-Frequency-Link-Based Electric Energy Routers*

Shanghai Jiao Tong University, Shanghai, China

Sept. 2016–Jun. 2020

B.S. in Electrical Engineering and Automation

- Thesis: *Ceaseless Modulation and Control Strategies of MMC-SST Under MVDC Fault*

📖 SELECTED PUBLICATIONS

Journal Papers

- **H. Wang, et al.**, “High-Frequency-Link-based Reactive-Power Optimal Control for Modular Multi Active Bridge Converter,” in *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, vol. 5, no. 3, pp. 1333–1337, July 2024.
- **H. Wang, et al.**, “Universal Phase-Shift Modulation Scheme and Efficiency Optimization for Modular Multiactive Bridge Converter,” in *IEEE Transactions on Industrial Electronics*, vol. 71, no. 7, pp. 7312–7321, July 2024.
- **H. Wang, et al.**, “Switching Characterization and Power Loss Optimization for Modular Multiactive Bridge Converter Under Common Phase Shift Control,” in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 11, no. 4, pp. 3924–3936, Aug. 2023.
- **H. Wang, et al.**, “ZVS Soft Switching Operation Region Analysis of Modular Multi Active Bridge Converter Under Single Phase Shift Control,” in *IEEE Transactions on Industrial Electronics*, vol. 70, no. 7, pp. 6865–6875, July 2023.
- J. Zheng[†], **H. Wang[†] et al.**, “Cognitive Digital Twins-Based Model Predictive Control for High-Frequency Power Converters,” in *IEEE Transactions on Industrial Electronics*, vol. 72, no. 12, pp. 13310–13321, Dec. 2025.
- J. Zheng[†], **H. Wang[†] et al.**, “Physics-Embedded Neural ODEs for Sim-to-Real Edge Digital Twins of Hybrid Power Electronics Systems,” in *IEEE Transactions on Industrial Electronics*, vol. 73, no. 6, pp. 8616–8627, June 2026.
- J. Zheng[†], **H. Wang[†] et al.**, “Neural Surrogate Solver for Efficient Edge Inference of Power Electronic Hybrid Dynamics,” in *IEEE Transactions on Industrial Electronics*, vol. 73, no. 6, pp. 9523–9528, June 2026.
- J. Zheng[†], **H. Wang[†] et al.**, “From Simulation to Control: A Digital Twin-Driven Automatic Synthesis Framework for Model Predictive Control in Power Converters,” in *IEEE Transactions on Power Electronics*.
- D. Mou, **H. Wang et al.**, “High-Efficiency Time-Division Multiplexing Modulation Technology for Modular Multiactive Bridge Converters,” in *IEEE Transactions on Industrial Electronics*, vol. 71, no. 6, pp. 5745–5754, June 2024.

[†]These authors contributed equally to this work as co-first authors.

Conference Papers

- **H. Wang, et al.**, “Resonant Method-based Fully Automated Core Loss Measurement System for Sub-MHz Magnetics With Switched Capacitor Sequence,” *2026 IEEE 27th Workshop on Control and Modeling for Power Electronics (COMPEL)*, Cambridge, UK.
- **H. Wang, et al.**, “Online Full ZVS Optimization for Modular Multi-Active Bridge Converter in MV PET,” *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, San Antonio, TX, USA, 2026.
- **H. Wang, et al.**, “Automatic Loss Measurement System for MHz Magnetics Using High-Frequency Conditioning Circuits,” *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, Knoxville, TN, USA, 2025, pp. 1–7.
- **H. Wang, et al.**, “High-Frequency Conditioning Circuits for Power-Related Information Extraction in Non-Sinusoidal Power Electronic Systems,” *2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Atlanta, GA, USA, 2025, pp. 2857–2864.
- **H. Wang, et al.**, “Model-Free Approximate Fundamental Reactive Power Minimization for Modular Multi-Active Bridge Converter,” *2024 IEEE 7th International Electrical and Energy Conference (CIEEC)*, Harbin, China, 2024, pp. 2068–2073.
- **H. Wang, et al.**, “Comparison and Improvement of ZVS Operation Under Different Modulation Strategies for Modular Multi Active Bridge Converters,” *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, Nashville, TN, USA, 2023, pp. 2556–2563.

MAIN RESEARCH EXPERIENCES

University of Texas at Austin, Austin, TX, USA

Project Lead at Power Electronics and Magnetics Group Advisor: Prof. Alex Hanson

Data-Driven Characterization of Magnetic Core Loss Jan. 2026–Present

- Developed a learnable symbolic sparse identification framework to discover explicit equations of core loss from measured data and physical knowledge, facilitating advanced magnetic characterization and design
- Submitted a paper to *IEEE TPEL* and a paper to *IEEE APEC 2027*

Automated Loss Measurement of Very-High-Frequency Power Magnetics Jun. 2024–Present

- Developed an automated system to measure core losses of MHz magnetics with great accuracy within seconds
- Published 2 papers in *IEEE COMPEL* and submitted a paper to *IEEE TPEL*

Columbia University, New York, NY, USA

Project Lead at Motor Drives and Power Electronics Lab Advisor: Prof. Matthias Preindl

Integrated Onboard AC-DC Electric Vehicle Charger Oct. 2023–Mar. 2024

- Designed an integrated onboard EV charger with only one conversion stage and a common division factor control to allow smooth power sharing and flexible power transfer

Tsinghua University, Beijing, China

Project Lead at High Power Electronics and Advanced Power Transmission Lab Advisor: Prof. Shiqi Ji

Reactive-Power Optimal Control for MMAB Converters May 2022–May 2023

- Derived a frequency-domain model of modular multi-active bridge (MMAB) converters, decoupled active and reactive powers, and proposed reactive-power optimal control to eliminate the fundamental reactive power
- Published a journal paper in *IEEE JESTIE* and a conference paper in *IEEE CIEEC 2024*

Universal Modulation and Global Optimization for MMAB Converters May 2022–Jan. 2023

- Proposed common phase-shift control and developed a local optimization strategy to minimize power losses and largely improve the system efficiency with a maximum increment of 16.55%

- Proposed a universal multi-phase-shift modulation scheme for MMAB converters and developed a global optimization strategy to improve the efficiency with a maximum increment of 20.49%
- Published a paper in *IEEE TIE*, a paper in *IEEE JESTPE*, and a conference paper in *IEEE ECCE 2023*

Zero-Voltage Switching Operation Analysis for MMAB Converters Nov. 2021–Jan. 2022

- Analyzed the operation and switching characterization of MMAB converters under phase-shift control, defined different ZVS regions, and generated an overall flow chart of discriminating ZVS operation
- Published a journal paper in *IEEE Transactions on Industrial Electronics*

WORK EXPERIENCES

Delta Electronics (Americas) Ltd., Durham, NC, USA May 2026–Aug. 2026

R&D Intern at Milan M. Jovanović Power Electronics Lab

Current-Limiting Strategies for Solid-State Transformer in Data Centers May 2026–Aug. 2026

- Proposed current-limiting strategies for both LLC and dual active bridge (DAB) converters of solid-state transformers in data centers to ensure uninterrupted power supply and improve the system reliability
- Submitted 2 journal papers to *IEEE TPEL* and one paper to *IEEE JESTPE*

HONORS

Best Poster Reward at IEEE COMPEL 2026 2026

Cockrell Engineering Fellowship at University of Texas at Austin 2024–2028

Wei Family Private Foundation Fellowship at Columbia University 2023

Outstanding Master Thesis at Tsinghua University (Top 1%) 2023

Outstanding Graduate of Beijing at Tsinghua University (Top 1%) 2023

East Cable Scholarship at Tsinghua University (Top 5%) 2022

Siyuan Electric Scholarship at Tsinghua University (Top 5%) 2021

PROFESSIONAL ACTIVITIES

Manuscript Reviewer Dec. 2022–Present

- IEEE Transactions on Power Electronics (IEEE TPEL)
- IEEE Transactions on Industrial Electronics (IEEE TIE)
- IEEE Journal of Emerging and Selected Topics in Power Electronics (IEEE JESTPE)
- IEEE Journal of Emerging and Selected Topics in Industrial Electronics (IEEE JESTIE)
- IEEE Transactions on Consumer Electronics (IEEE TCE)
- IEEE Transactions on Power Delivery (IEEE TPWRD)
- IEEE Open Journal of Power Electronics (IEEE OJPEL)
- IEEE Open Journal of the Industrial Electronics Society (IEEE OJIES)
- IEEE Energy Conversion Conference and Exposition (IEEE ECCE)
- IEEE Applied Power Electronics Conference (IEEE APEC)

MISCELLANEOUS

- **Programming:** Python, C/C++ and Verilog HDL, MATLAB, LaTeX, etc.
- **Proficient Software:** MATLAB, Anaconda, PyTorch, Altium Designer, KiCad, Code Composer Studio, Quartus, Xilinx, TCAD, Cadence
- **Language:** Mandarin (native), English (fluent), Japanese (basic)
- **Interest:** Mountaineering, Calisthenics, Powerlifting, Rock Climbing, Running, Photographing, etc.